

CSA0704-COMPUTER NETWORKS
ASSESSMENT TOOL 3 | TECHNICAL PRESENTATION

Topic - Edge Computing and Reliable Data Transmission

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THE CONCEPT

What Is Edge Computing?

- Edge computing pushes data processing away from a centralized cloud and toward the network edge — close to where data is generated. It directly supports OSI Layers 1–4 by cutting transmission distance, reducing congestion, and improving delivery reliability.



Lower Latency

Processing near the source shortens the round trip across Physical and Data Link layers.



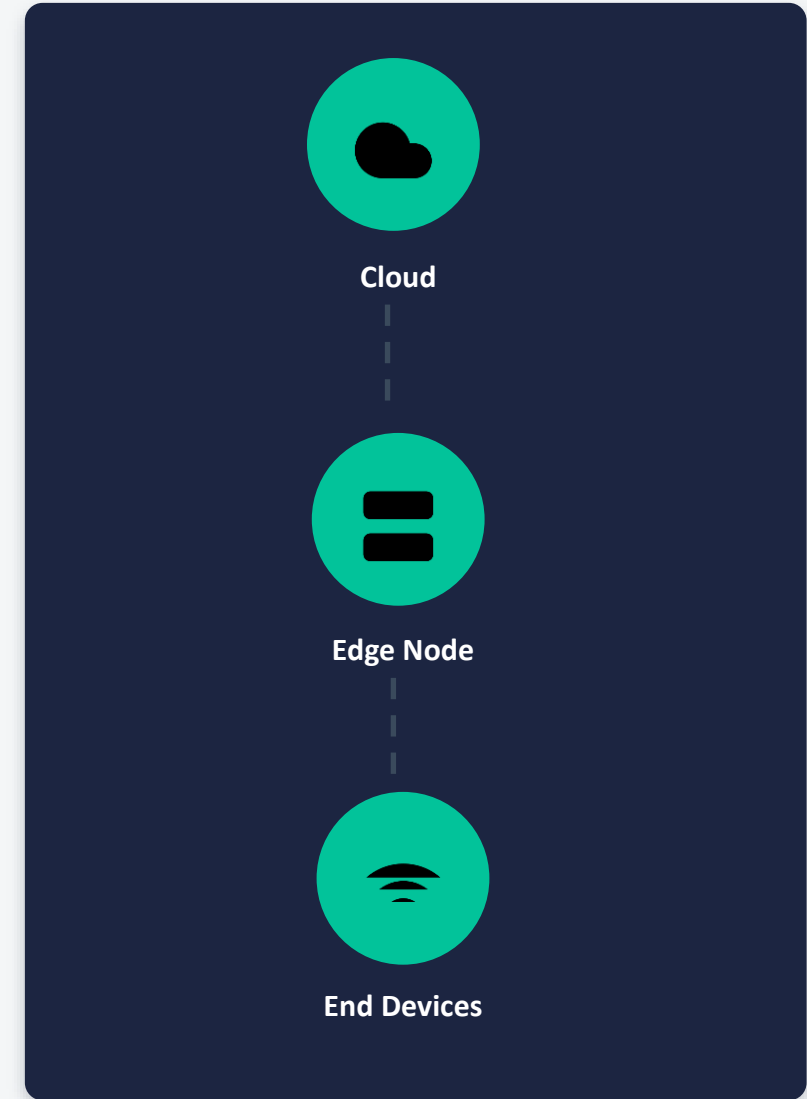
Local Processing

Edge nodes handle filtering and computation before data reaches the core network.



Reduced Congestion

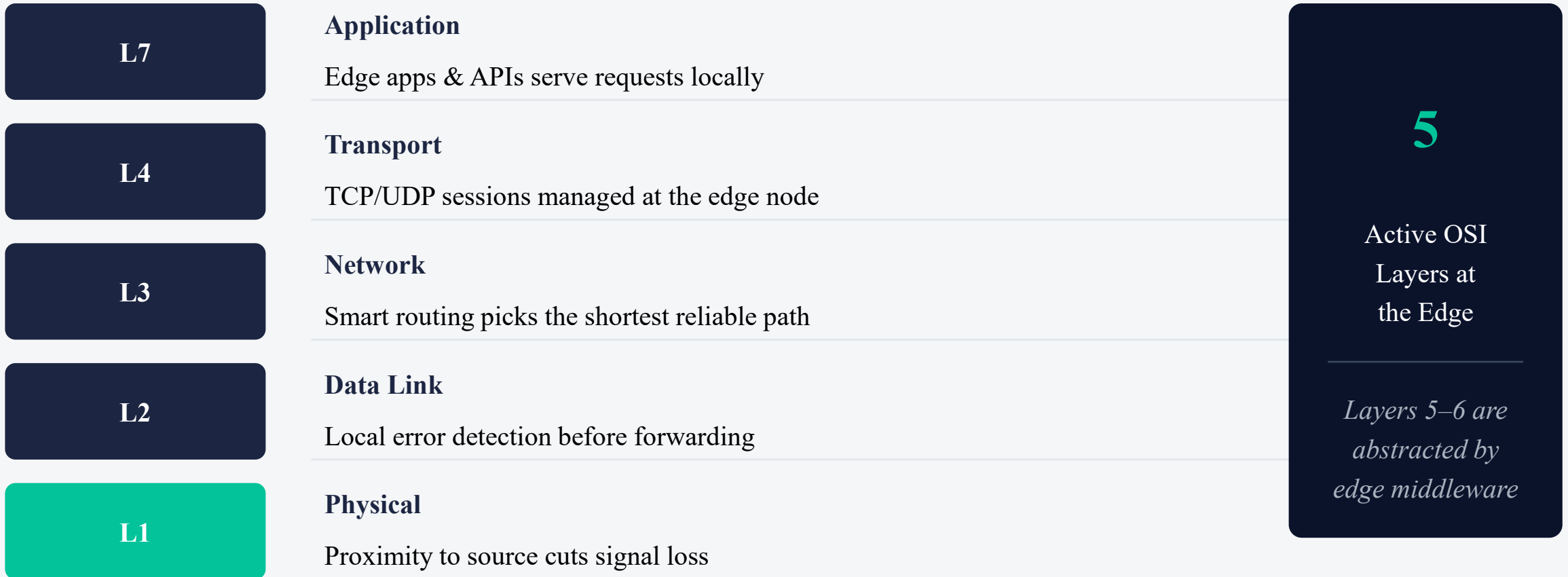
Less traffic on backbone links means fewer collisions and retransmissions.



ARCHITECTURE

Edge Computing Mapped to the OSI Model

Each OSI layer plays a distinct role in how an edge node receives, verifies, and forwards data reliably.



RELIABILITY

How Edge Computing Keeps Transmission Reliable



Local Retransmission

Edge nodes detect lost packets and retransmit within milliseconds, instead of round-tripping to a distant cloud server.



Error Checking Nearby

Checksums and CRC validation happen at the closest hop, catching Layer 2 errors before they propagate further.



Adaptive Routing

Network-layer intelligence reroutes traffic around congestion or failure, keeping delivery paths reliable.



Reduced Jitter

Shorter physical paths mean more consistent timing, which is critical for real-time transport-layer sessions.

TAKEAWAYS

Benefits, Challenges & the Road Ahead

BENEFITS

- Faster, more reliable delivery across all OSI layers
- Lower bandwidth cost on core network links
- Better resilience during partial network failure

CHALLENGES

- Managing security across many distributed nodes
- Coordinating consistency between edge and cloud
- Higher deployment and maintenance complexity

CONCLUSION

Edge computing reinforces the OSI model rather than replacing it — by executing each layer's function closer to the source, it delivers the low-latency, high-reliability transmission that next-generation networks demand.



Thank You